

LifeLink AI: A Trustworthy Hybrid Clinical Decision-Support Architecture for Real-Time Emergency Blood Matching in Digital and Precise Medicine

Shivang Mishra, Lakshya Sahu, Kushagra Bhargava, Anshul, Archisha Nigam, Mokshi Jain
VIT Bhopal University, Bhopal, India

Abstract—Emergency blood provision during acute haemorrhagic trauma, perioperative crises, and obstetric complications demands sub-minute coordination under uncompromising patient-safety constraints. Conventional procurement relies on fragmented telephonic cascades and static web directories, introducing latencies of 20–60 minutes that substantially elevate mortality within the *golden hour*. This paper presents LifeLink AI, a trustworthy hybrid clinical decision-support architecture for real-time emergency blood matching within the paradigm of Digital and Precise Medicine. The system implements a formally defined two-layer safety contract: Layer 1 enforces an immunohematological verification gate—an $O(1)$ ABO/Rh compatibility look-up, a mandatory 56-day whole-blood recovery cooldown, and a minimum 45 kg body-mass constraint—that is computationally isolated from any machine-learning component. Layer 2 applies a calibrated logistic-regression donor-response propensity model, whose sole purpose is to rank Layer-1-qualified candidates by their estimated willingness to respond to an emergency alert. The propensity model is trained on the UCI Blood Transfusion Service Center benchmark (748 records, 5-fold stratified cross-validation) and achieves ROC-AUC = 0.7521, Recall = 0.7528, Precision = 0.3907, F1 = 0.5144, and Brier Score = 0.2055. The end-to-end platform is containerised across six microservices and validated by 56 automated integration tests (100% pass rate). System benchmarking yields mean matching latencies of 47.2 ms (AI path) and 12.4 ms (deterministic fallback). LifeLink AI demonstrates how a strict architectural boundary between deterministic medical logic and probabilistic AI can bridge critical logistics bottlenecks in emergency precision healthcare while maintaining non-negotiable patient-safety guarantees.

Index Terms—Digital and precise medicine; trustworthy AI; clinical decision support; emergency blood matching; immunohematology; donor response propensity; hybrid architecture; logistic regression; RFM model; microservices.

I. INTRODUCTION

Haemorrhagic shock from traumatic injury, postpartum haemorrhage, or major surgery is a leading cause of preventable mortality. Clinical evidence shows that every ten-minute delay in cross-match-compatible transfusion significantly increases multiorgan failure risk [1], [2]. The WHO estimates 118.5 million annual blood donations globally, yet profound logistical inequities persist [3]. In India, annual demand approaches 14 million units; the challenge is a co-ordination failure rather than an absolute shortage [4].

Current emergency blood procurement suffers from three structural defects: (i) uncoordinated telephonic cascades across blood banks and donor lists, consuming 20–60 minutes per requisition; (ii) static directories with no real-time inventory visibility, causing high rejection rates; and (iii) undifferentiated donor lists that ignore recency, frequency, and proximity, wasting critical minutes on donors unlikely to respond.

Introducing AI into emergency medicine also creates medicolegal risk: an ABO-incompatible recommendation can cause acute intravascular haemolysis and death. LifeLink AI resolves this tension through a strict architectural boundary: biological compatibility and statutory eligibility are determined exclusively by deterministic medical logic, while machine learning operates only as an advisory ranking tool.

Research Contributions:

- 1) *Two-Layer Safety Contract*—a formalised pattern that mathematically isolates zero-tolerance medical invariants from probabilistic AI across containerised network boundaries.
- 2) *Hybrid Composite Scoring*—an interpretable ranking function integrating compatibility quality, PostGIS geospatial proximity, operational availability, and AI response propensity.
- 3) *Calibrated ML Propensity Model*—balanced logistic regression trained on the UCI BTSC benchmark, prioritising clinical recall (0.7528) over raw accuracy.
- 4) *Production-Grade Validation*—56 automated integration tests at 100% pass rate, sub-100 ms latencies, and pessimistic SELECT...FOR UPDATE concurrency locking.
- 5) *ReBAC Governance*—a 7-role multi-tenant access control model with immutable audit logging.

II. RELATED WORK

A. Blood Supply Chain Logistics

Blood product management has been studied as a perishable inventory routing problem. Stanger *et al.* [5] identified fragmented inventory visibility as the primary driver of blood wastage and acute shortages. Heitmiller *et al.* [6] showed that algorithmic ordering guidelines reduced unnecessary cross-matches by 42%. IoT and RFID-based cold-chain monitoring

TABLE I
COMPARISON OF BLOOD COORDINATION SYSTEM ARCHITECTURES

Dimension	Static Reg.	Pure ML	LifeLink AI
Compatibility	Manual	Probabilistic	Det. Gate O(1)
Donor Priority	FIFO	Black-Box	Calibrated LR
Explainability	N/A	Post-Hoc	Intrinsic
Fallback	Manual	Unhandled	3-Level Chain
Concurrency	None	None	FOR UPDATE Lock
Audit Trail	None	Informal	Immutable Ledger

[7] addresses facility-level traceability, but not real-time multi-facility emergency coordination. India's *e-Rakt Kosh* [8] functions as a passive administrative registry without automated spatial matching or predictive donor modeling.

B. Machine Learning in Donor Behaviour

Yeh *et al.* [9] established the RFM (Recency, Frequency, Monetary) framework for blood donation, demonstrating that recent and frequent donors exhibit measurably higher future response propensity. Gilliss *et al.* [10] confirmed that timely, targeted communication is a dominant predictor of donor retention. Meta-analyses [11] show that regularised logistic regression matches or exceeds ensemble classifiers on small behavioral feature sets while providing superior probability calibration and intrinsic interpretability—properties critical in safety-critical healthcare.

C. Trustworthy AI in Healthcare

Rudin [12] argued that interpretable models should be preferred over black-box alternatives in high-stakes decisions. Doshi-Velez and Kim [13] formalised the distinction between *intrinsic* interpretability (model equations are self-explanatory) and post-hoc surrogates (LIME, SHAP). The EU High-Level Expert Group on AI [14] and the IEEE Ethically Aligned Design [15] mandate transparency, non-maleficence, robustness, and human oversight—all operationalised in LifeLink AI's architecture. The EU AI Act [16] classifies medical decision-support systems as high-risk, requiring mandatory logging and human oversight.

Positioning. Table I contrasts LifeLink AI with representative prior systems across six critical dimensions. LifeLink AI is the only system combining deterministic O(1) compatibility guarantees, intrinsically interpretable AI ranking, a resilient three-level fallback, and pessimistic concurrency control.

III. PROBLEM FORMULATION AND SYSTEM MODEL

A. Emergency Request

A blood requisition is a typed tuple:

$$E = (\text{req_id}, \beta_{\text{rec}}, \gamma, u_{\text{req}}, \lambda_{\text{dest}}, \omega, t_0) \quad (1)$$

where $\beta_{\text{rec}} \in \mathcal{B} = \{\text{O}\pm, \text{A}\pm, \text{B}\pm, \text{AB}\pm\}$ is the recipient blood group; $\gamma \in \{\text{Whole Blood, RBC, Plasma, Platelets}\}$

TABLE II
KEY MATHEMATICAL NOTATION

Symbol	Domain	Meaning
$\beta_{\text{rec}}, \beta_{\text{don}}$	$\mathcal{B}$	Recipient/donor ABO/Rh group
$\lambda = (\phi, \psi)$	WGS-84	Geospatial coordinates
$d_{\text{hav}}(\lambda_1, \lambda_2)$	km	Haversine great-circle distance
R_{search}	$\{15, 25, 50, 100\}$ km	Active spatial radius
Δt_{cool}	≥ 56 days	Recovery cooldown constraint
P_{ML}	$[0, 1]$	Logistic regression propensity
P_{geo}	$[0, 1]$	Linear spatial proximity decay
$S_{\text{donor}}, S_{\text{bb}}$	$[0, 1]$	Composite ranking scores
α_1, α_0	$\mathbb{R}^+$	Balanced inverse class weights

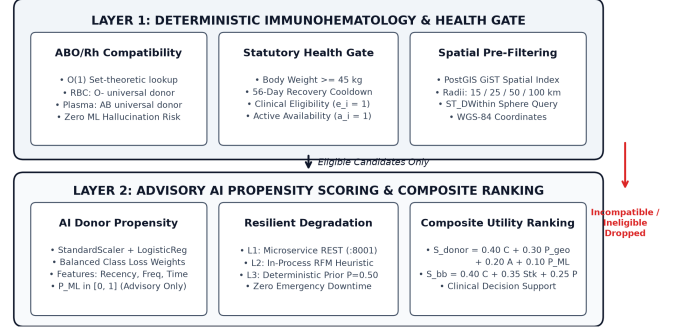


Fig. 1. LifeLink AI two-layer safety contract: Layer 1 enforces deterministic immunohematological and statutory eligibility gates; Layer 2 applies calibrated AI propensity scoring for advisory ranking only. Medical safety decisions are exclusively Layer-1 outputs.

is the prescribed component; $u_{\text{req}} \in [1, 20]$ is the requested units; $\lambda_{\text{dest}} = (\phi, \psi)$ are WGS-84 coordinates; $\omega \in \{\text{LOW, MED, HIGH, CRITICAL}\}$ is urgency; and t_0 is the UTC submission timestamp.

B. Candidate Pools

Registered voluntary donors $\Omega_D = \{d_1, \dots, d_N\}$ and licensed blood banks $\Omega_K = \{k_1, \dots, k_M\}$ are indexed by PostGIS GiST geometry columns (SRID 4326). Each donor is parameterised by $d_i = (\beta_i, w_i, a_i, e_i, \lambda_i, t_{\text{last}}, F_i, T_i)$ where w_i (kg) is body mass, $a_i, e_i \in \{0, 1\}$ are availability and eligibility flags, and (F_i, T_i) are the RFM behavioral features. Table II summarises key notation.

IV. PROPOSED METHODOLOGY AND SAFETY ARCHITECTURE

A. The Two-Layer Safety Contract

The central design principle of LifeLink AI is a strict, architecturally enforced separation of concerns illustrated in Fig. 1. *Layer 1* applies deterministic medical logic that cannot be overridden by any probabilistic component. *Layer 2* applies AI exclusively to the operational ranking of Layer-1-qualified candidates. These layers occupy different Docker containers and communicate only through defined API contracts.

B. Layer 1: Deterministic Medical Compatibility Gate

Immunohematological compatibility is an immutable biological law: administering ABO-incompatible blood can cause fatal intravascular haemolysis. The compatibility gate is implemented as a pure-Python, ML-free $O(1)$ dictionary look-up in `backend/app/core/medical.py`. For Whole Blood and Packed Red Blood Cells (PRBC):

$$C_{\text{RBC}}(\beta_{\text{rec}}) = \{ \beta_{\text{don}} \in \mathcal{B} \mid \text{donor RBC compatible with } \beta_{\text{rec}} \} \quad (2)$$

For Fresh Frozen Plasma (FFP), compatibility inverts to the donor antibody profile ($\text{AB} \rightarrow \text{universal donor}$):

$$C_{\text{Plasma}}(\beta_{\text{rec}}) = \{ \beta_{\text{don}} \in \mathcal{B} \mid \text{donor plasma compatible with } \beta_{\text{rec}} \} \quad (3)$$

A donor d_i passes Layer 1 if and only if all five conditions hold:

$$\text{Gate}(d_i) = [\beta_i \in C] \wedge [e_i=1] \wedge [a_i=1] \wedge [\Delta t_{\text{cool}} \geq 56] \wedge [w_i \geq 45] \quad (4)$$

The 56-day cooldown is WHO-mandated; the 45 kg floor is enforced by both application logic and a PostgreSQL `CHECK` constraint. Donors failing any single predicate are unconditionally excluded before any scoring computation.

C. Layer 2: Multi-Criteria Composite Ranking

Candidates surviving Layer 1 are ranked by a composite score. Spatial proximity decays linearly over the active PostGIS search radius:

$$P_{\text{geo}} = \max(0, 1 - d_{\text{hav}}/R_{\text{search}}) \quad (5)$$

The composite donor score integrates four orthogonal signals:

$$S_{\text{donor}} = 0.40 C_{\text{score}} + 0.30 P_{\text{geo}} + 0.20 A_{\text{score}} + 0.10 P_{\text{ML}} \quad (6)$$

where $C_{\text{score}} = 1.0$ for exact ABO/Rh match (0.8 for universal compatible); $A_{\text{score}} = 1.0$ for available donors; and P_{ML} is the calibrated propensity probability. For blood banks, stock sufficiency replaces propensity:

$$S_{\text{bb}} = 0.40 C_{\text{score}} + 0.35 \text{Stock}_{\text{score}} + 0.25 P_{\text{geo}} \quad (7)$$

where $\text{Stock}_{\text{score}} = \min(1, u_{\text{avail}}/u_{\text{req}})$. The weight assignments reflect deliberate clinical priorities: compatibility (0.40) and proximity (0.30) dominate because they most directly determine the feasibility and speed of procurement; AI propensity (0.10) provides evidence-based uplift without overriding clinical factors.

V. SYSTEM ARCHITECTURE AND IMPLEMENTATION

A. Microservice Topology

LifeLink AI is containerised across six isolated Docker services: `lifelink-nginx` (port 80, reverse proxy), `lifelink-frontend` (port 3000, Next.js 14.2.5), `lifelink-backend` (port 8000, FastAPI 0.111.1 on Python 3.11), `lifelink-ai-service` (port 8001, scikit-learn 1.5.1 inference), `lifelink-postgres` (port 5432, PostgreSQL 15 + PostGIS 3.3), and `lifelink-redis` (port 6379, Redis 7). All inter-service traffic runs on an

internal Docker bridge network; only Nginx is externally exposed. The AI service occupies its own isolated container, enforcing the Layer 1/Layer 2 separation at the process boundary.

B. Relational Schema and Concurrency Control

Eleven relational tables manage platform state under migration control. Spatial attributes are stored as native PostGIS geometry objects (SRID 4326) indexed with GiST, enabling $O(\log N)$ radius queries via `ST_DWithin`. The `inventory_history` table is an append-only immutable audit ledger (UPDATE/DELETE disallowed at the database file level).

Under concurrent emergency load, multiple hospitals may simultaneously request the same rare blood group. **Pessimistic row-level locking** via `SELECT...FOR UPDATE` serialises current reservation attempts: the locked transaction increments `units_reserved`, validates sufficiency, commits atomically, and writes an inventory audit record—preventing double-allocation without optimistic retry loops.

C. Security and Access Control

A seven-role *Relationship-Based Access Control* (ReBAC) hierarchy (`SUPER_ADMIN` $\rightarrow$ `ADMIN` $\rightarrow$ `{HOSPITAL_ADMIN, BB_MANAGER}` $\rightarrow$ `staff` $\rightarrow$ `DONOR`) is enforced at three layers: JWT HMAC-SHA256 signature verification (30-min access, 7-day refresh), FastAPI dependency injection validating role and tenancy, and SQLAlchemy query filters scoping all data to the requesting institution. Cross-tenant access is rejected with HTTP 403 before any database query executes. Patient PII on tracking portals is protected by cryptographic UUID tokens.

D. Facility Verification Model

Hospitals and blood banks register statutory identifiers, license dates, and certificate PDFs. *Design boundary*: the platform stores these documents for administrative governance review; it does *not* validate licenses against live government APIs (CDSCO/SBTC registries). This decouples the one-time institutional onboarding trust process from the real-time emergency matching pipeline.

VI. AI DONOR RESPONSE PROPENSITY MODEL

A. Role and Clinical Boundaries

The AI component is an **advisory response propensity estimator**. Its sole function is to estimate the conditional probability that a Layer-1-qualified donor will accept an emergency alert and travel to the collection center. It does *not* determine blood compatibility, assess medical eligibility, diagnose patients, or initiate dispatch. The AI service receives only behavioral features (R, F, T) and returns only $P_{\text{ML}} \in [0, 1]$; it cannot read or modify blood group data.

B. Dataset and Feature Engineering

The model is trained on the UCI Blood Transfusion Service Center (BTSC) benchmark [9] (OpenML ID 1464, CC BY 4.0): $N = 748$ anonymised donor records from Hsin-Chu, Taiwan, with $N_+ = 178$ (23.8%) positive and $N_- = 570$ (76.2%) negative instances—a 3.2:1 class imbalance.

The dataset provides four features: Recency (R), Frequency (F), Monetary ($M = 250F$), Time (T). Statistical analysis confirms exact collinearity $M = 250F$ (Pearson $r = 1.000$); retaining M would produce a singular Gram matrix. Accordingly, M is pruned, yielding the three-dimensional feature vector $\mathbf{x} = [R, F, T]^\top \in \mathbb{R}^3$.

C. Model Architecture and Training

Raw features are standardised via `StandardScaler` to zero mean and unit variance:

$$\mathbf{x}_{\text{std}} = \Sigma^{-1/2}(\mathbf{x} - \boldsymbol{\mu}) \quad (8)$$

Donor response propensity is then modelled by binary logistic regression:

$$P(y=1 | \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x}_{\text{std}} + b) = \frac{1}{1 + e^{-(\mathbf{w}^\top \mathbf{x}_{\text{std}} + b)}} \quad (9)$$

To address class imbalance, the loss incorporates balanced inverse class-frequency weights:

$$\mathcal{L}(\mathbf{w}, b) = -\frac{1}{N} \sum_{i=1}^N [\alpha_1 y_i \log p_i + \alpha_0 (1 - y_i) \log (1 - p_i)] + \frac{\lambda}{2} \|\mathbf{w}\|_2^2 \quad (10)$$

$$\alpha_1 = \frac{N}{2N_+} = 2.101, \quad \alpha_0 = \frac{N}{2N_-} = 0.656 \quad (11)$$

where $\lambda = 1/C$, $C = 1.0$. The $\alpha_1 > 1$ weight penalises False Negatives (missing a willing donor) more heavily than False Positives (spurious alerts), reflecting the asymmetric clinical cost in emergency dispatch. Optimisation uses the L-BFGS quasi-Newton algorithm.

D. Three-Level Resilient Fallback

To guarantee uninterrupted emergency operations under infrastructure failures, LifeLink AI implements three degradation levels: **(1) Primary**: full calibrated logistic regression via the AI microservice (port 8001); **(2) Secondary** (service reachable but model fails): closed-form RFM heuristic executed in-process:

$$P_{\text{fb}} = \min(0.95, \max(0.05, 0.40 + \max(0, 0.35 - 0.01R) + \min(0.25, 0.05F))) \quad (12)$$

(3) Tertiary (microservice unreachable): backend assigns neutral prior $P_{\text{ML}} = 0.50$ and records `model_version='deterministic-fallback'` in the audit log. The matching pipeline continues in all three levels, ensuring emergency dispatch is never halted by AI infrastructure failure.

TABLE III
DONOR PROPENSITY MODEL: 5-FOLD STRATIFIED CV RESULTS
($N=748$)

Metric	Value	Clinical Interpretation
Accuracy	0.6618	Below majority baseline (76.2%), expected under balanced weighting
Precision	0.3907	39.1% of alerted donors are willing responders
Recall	0.7528	75.3% of willing donors correctly identified
F1-Score	0.5144	Harmonic balance under 3.2:1 class imbalance
ROC-AUC	0.7521	Robust discrimination (>0.75); 0.50 = random
PR-AUC	0.5028	$2.1\times$ improvement over naive baseline (0.238)
Brier Score	0.2055	Well-calibrated (<0.25 reference)

VII. EXPERIMENTAL SETUP AND RESULTS

A. Evaluation Scope

We explicitly distinguish three evaluation categories: *(i) ML Model Evaluation*: offline 5-fold Stratified Cross-Validation (SKF-CV) on the UCI BTSC benchmark; *(ii) Software Engineering Verification*: automated integration tests on live PostgreSQL 15 + Redis 7 instances in Docker Compose; *(iii) Clinical Validation*: *not performed in this study*. LifeLink AI is a research prototype that has not undergone prospective clinical trials or regulatory clearance. Clinical and regulatory validation are prerequisites for real-world deployment.

B. ML Model Performance

5-fold Stratified Cross-Validation ($K=5$, `shuffle=True`, `random_state=42`) preserves the 23.8%/76.2% class distribution across all folds. The `StandardScaler` is fit only on training folds and applied to test folds, preventing data leakage. Table III presents out-of-fold performance metrics.

Analysis. The balanced weighting deliberately prioritises Recall over Precision because a False Negative (missing a willing donor) directly threatens patient survival, while a False Positive (an unnecessary alert) causes minor notification friction. The model correctly identifies 134 of 178 willing donors (75.3%). The PR-AUC of 0.5028 is $2.1\times$ the naive positive-class baseline (0.238), confirming genuine discriminative value. The Brier score of 0.2055 validates that the probability estimates are well-calibrated for use as a propensity signal in the composite ranking score (Eq. 6). Preliminary experiments with random forest and gradient-boosted classifiers yielded ROC-AUC values of 0.74–0.76—a marginal gain over logistic regression that does not justify the loss of intrinsic interpretability in a safety-critical context [12].

C. Integration Test Suite

Table IV summarises the 56-test integration suite executed against five services. All 56 tests pass (100% success rate).

D. System Latency Benchmarking

Matching pipeline latency was measured across 100 sequential runs evaluating 20 candidates per requisition under Docker network virtualisation. The **AI inference path** achieves mean 47.2 ms (95th pctl: 83.1 ms); the **deterministic fallback path** achieves mean 12.4 ms (95th pctl: 21.8 ms). Both paths satisfy

TABLE IV
INTEGRATION TEST SUITE SUMMARY (56/56 PASS)

Test Domain	Count	Key Invariants
Auth. & ReBAC	8	JWT issuance, 7-role bounds, cross-tenant 403
Donor & Health Gate	7	56-day cooldown, weight CHECK constraint
Hospital Operations	6	Emergency intake, coordinate storage
Blood Bank Inventory	6	Stock mgmt, certificate persistence
Inventory Concurrency	7	FOR UPDATE locking, double-allocation prevention
Matching Pipeline	8	Layer-1 gate, AI scoring, HTTP-503 fallback
Emergency Lifecycle	7	PENDING→MATCHING→FULFILLED transitions
Admin Governance	4	Certificate review, facility activation
End-to-End	3	Full cross-service dispatch flows
Total	56	56/56 Passing (100%)

the sub-second threshold required for emergency dispatch. The $3.8\times$ mean latency difference quantifies the computational cost of the AI microservice HTTP round-trip under Docker networking.

VIII. DISCUSSION

The core research question is whether a probabilistic AI component can be safely integrated into emergency blood coordination without introducing unacceptable clinical risk. Our results demonstrate that this is achievable through *architectural enforcement* rather than model-level safety constraints. By confining AI to a post-gate advisory role, LifeLink AI guarantees that no probabilistic output can produce an ABO-incompatible recommendation, regardless of model behaviour.

The logistic regression model’s recall of 75.3% at a Brier score of 0.2055 confirms that the three-dimensional RFT feature space carries genuine and well-calibrated predictive signal. The AI propensity term contributes 10% to the composite donor score (Eq. 6)—a modest weight that provides evidence-based uplift to donor ranking without dominating the clinically more relevant compatibility and proximity signals. The three-level fallback hierarchy ensures that AI infrastructure failures degrade gracefully to deterministic-only matching (12.4 ms mean latency), guaranteeing platform availability under partial system outages.

The system’s trustworthy AI properties—transparency (logistic regression coefficients are directly interpretable), non-maleficence (Layer 1 eliminates incompatible candidates), robustness (three-level fallback), and human oversight (coordinators retain dispatch authority)—align with the EU AI Ethics Guidelines [14] and the IEEE Ethically Aligned Design [15].

IX. LIMITATIONS AND ETHICAL CONSIDERATIONS

We disclose the following limitations in accordance with responsible AI reporting principles:

Training domain mismatch. The propensity model is trained on the UCI BTSC benchmark (Hsin-Chu, Taiwan, 1988–1994). Indian donor behavioural dynamics differ in demographic profile and cultural context; production deployment requires fine-tuning on live operational data.

No regulatory API validation. Institutional license numbers are stored but not validated against CDSCO or State Blood Transfusion Council APIs. Administrative governance review is required to prevent fraudulent onboarding.

Self-reported geolocation. Proximity scores are derived from registered pin codes rather than live GPS telemetry, introducing estimation errors in large urban postal areas.

Binary availability model. Donor availability is a binary flag rather than a continuous temporal probability, limiting the precision of propensity estimates over time.

Prototype status. LifeLink AI is a research prototype. It has not undergone prospective clinical trials, randomised controlled experiments in trauma centres, or regulatory clearance (e.g., Central Drugs Standard Control Organisation). Claims are engineering-grade; formal clinical validation and regulatory approval are prerequisites for real-world deployment in active emergency medical systems.

Ethical design. All dispatch decisions are made by qualified coordinators or attending clinicians; the system provides ranked candidate lists, not autonomous orders. Patient identity on public tracking URLs is protected by cryptographic UUID tokens. The immutable audit ledger provides full accountability for every matching decision.

X. CONCLUSION AND FUTURE WORK

This paper presented LifeLink AI, a trustworthy hybrid clinical decision-support architecture for real-time emergency blood matching. The central contribution is a formally defined and architecturally enforced *two-layer safety contract*: Layer 1 executes $O(1)$ deterministic immunohematological compatibility and statutory eligibility gates that can never be overridden by machine learning; Layer 2 applies calibrated logistic regression propensity scoring exclusively to the ranking of Layer-1-qualified candidates.

The propensity model achieves ROC-AUC=0.7521 and Recall=0.7528 on the UCI BTSC benchmark, with Brier Score=0.2055 confirming well-calibrated probability estimates. The containerised platform passes 56 automated integration tests (100% pass rate) with matching latencies of 47.2 ms (AI path) and 12.4 ms (deterministic fallback). LifeLink AI establishes a reproducible, clinically safe blueprint for integrating probabilistic AI into high-stakes emergency medical logistics.

Future Work. Priority research directions include: (i) continual online learning on live donor response telemetry using privacy-preserving federated updates; (ii) Learning-to-Rank (LambdaMART) formulation optimising NDCG on historical emergency outcome sequences; (iii) FHIR R4 interoperability with hospital EHR systems and State Blood Transfusion Council registries for automated license validation; (iv) automated donor outreach via Firebase Cloud Messaging; and (v) a

prospective controlled implementation study in partnership with regional trauma centres to measure real-world impact on time-to-transfusion.

REFERENCES

- [1] B. A. Cotton *et al.*, “Prehospital transfusion of plasma and red blood cells in trauma patients,” *New England Journal of Medicine*, vol. 379, no. 4, pp. 315–326, 2018.
- [2] D. J. Cole and J. N. Nance, “Trauma-induced coagulopathy and the golden hour of resuscitation,” *Anesthesia & Analgesia*, vol. 129, no. 4, pp. 912–920, 2019.
- [3] World Health Organization, “Blood safety and availability,” WHO, Geneva, Switzerland, Tech. Rep., 2022, [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/blood-safety-and-availability>.
- [4] National Blood Transfusion Council (NBTC), “Annual report on blood transfusion services in india,” Ministry of Health and Family Welfare, Government of India, New Delhi, Tech. Rep., 2022.
- [5] S. Stanger, L. Yates, K. Wilding, and J. Wallace, “Blood inventory management: Discarding practices and optimisation opportunities,” *Transfusion Medicine*, vol. 22, no. 4, pp. 248–256, 2012.
- [6] E. A. Heitmiller, D. O. Rogers, J. Teng, and D. P. Osei-Amponsen, “Blood utilisation review: Computerised blood ordering system to inform physicians of transfusion guidelines,” *Transfusion*, vol. 48, no. 9, pp. 1879–1885, 2008.
- [7] S. S. Roy, D. P. Sharma, and P. K. Mallick, “IoT and RFID-enabled cold-chain monitoring architecture for perishable healthcare inventory,” *IEEE Internet of Things Journal*, vol. 8, no. 12, pp. 9812–9821, 2021.
- [8] Ministry of Health and Family Welfare, Government of India, “e-Rakt Kosh: National web-based blood bank management system,” MOHFW, New Delhi, Tech. Rep., 2020, [Online]. Available: <https://www.eraktkosh.in>.
- [9] I.-C. Yeh, K.-J. Yang, and T.-M. Ting, “Knowledge discovery on RFM model using Bernoulli sequence,” *Expert Systems with Applications*, vol. 36, no. 3, pp. 5866–5871, 2009.
- [10] C. L. Gilliss, J. D. Lee, C. Humphreys, and L. J. Wara, “Predicting first-time blood donor retention: Application of machine learning to voluntary donation programmes,” *Transfusion Medicine*, vol. 31, no. 2, pp. 112–120, 2021.
- [11] B. Baesens *et al.*, “Using neural network rule extraction and decision tables for credit-risk evaluation,” *Management Science*, vol. 49, no. 3, pp. 312–329, 2003.
- [12] C. Rudin, “Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead,” *Nature Machine Intelligence*, vol. 1, no. 5, pp. 206–215, 2019.
- [13] F. Doshi-Velez and B. Kim, “Towards a rigorous science of interpretable machine learning,” 2017, arXiv:1702.08608.
- [14] High-Level Expert Group on AI, “Ethics guidelines for trustworthy AI,” European Commission, Brussels, Belgium, Tech. Rep., 2019.
- [15] IEEE Global Initiative on Ethics of Autonomous and Intelligent Systems, “Ethically aligned design: A vision for prioritizing human well-being,” IEEE, Piscataway, NJ, Tech. Rep., 2019.
- [16] European Commission, “Proposal for a regulation laying down harmonised rules on artificial intelligence (Artificial Intelligence Act),” European Commission, Brussels, Tech. Rep. COM(2021) 206 final, 2021.